WORKED SOLUTION

STEP III 1987, Question 2

Pure • Trigonometric Reduction Formulae

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/87-s3-q2

The key idea

The reduction formula comes from differentiating the "boundary term" $\cos^m x \cos nx$ and then using a compound-angle identity to turn $\sin x \cos nx$ into something involving $\sin(n-1)x$. Once we have it, both parts are about evaluating that boundary term at the limits: at 0 and π it either cancels or survives depending on parity, and at $\frac{\pi}{2}$ the factor $\cos^m x$ kills it.

(The printed formula has $-\cos^m \cos nx$; it should read $-\cos^m x \cos nx$.)

Solution

The reduction formula

Differentiate $\cos^m x \cos nx$ by the product rule:

$$\frac{\mathrm{d}}{\mathrm{d}x}(\cos^m x \cos nx) = -m\cos^{m-1} x \sin x \cos nx - n\cos^m x \sin nx.$$

The first term is not yet in the form we want. The compound-angle formula

$\sin(n-1)x = \sin nx \cos x - \cos nx \sin x$ gives $\sin x \cos nx = \cos x \sin nx - \sin(n-1)x$.

Substituting,

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}x}(\cos^m x \cos nx) &= -m\cos^m x \sin nx + m\cos^{m-1} x \sin(n-1)x - n\cos^m x \sin nx \\ &= -(m+n)\cos^m x \sin nx + m\cos^{m-1} x \sin(n-1)x.\end{aligned}$$

Integrating both sides (this needs $m \geq 1$ and $n \geq 1$ so that $m-1$, $n-1$ are non-negative),

$$\cos^m x \cos nx = -(m+n)I_{m,n} + mI_{m-1,n-1},$$

that is,

$$(m+n)I_{m,n} = -\cos^m x \cos nx + mI_{m-1,n-1},$$

as required (with the usual arbitrary constant absorbed into the indefinite integrals).

Part (i)

Write $J_{m,n} = \int_0^\pi \cos^m x \sin nx \, \mathrm{d}x$. Applying the formula between 0 and π, and using $\cos^m \pi \cos n\pi = (-1)^m(-1)^n$, for $m, n \geq 1$:

$$(m+n)J_{m,n} = 1 - (-1)^{m+n} + mJ_{m-1,n-1}.$$

If m and n have the same parity then $m+n$ is even, the boundary term vanishes and

$$J_{m,n} = \frac{m}{m+n} J_{m-1,n-1}.$$

Moreover $m-1$ and $n-1$ again have the same parity, so we may keep going.

Ask yourself Where does this chain of reductions stop, and why is the value there zero?

Each step lowers both indices by 1, so after $\min(m,n)$ steps one index is 0 and the other, having the same parity, is even. So it is enough to show $J_{k,0} = 0$ and $J_{0,k} = 0$ for every even $k \geq 0$.

- $J_{k,0} = \int_0^\pi \cos^k x \sin 0 \, \mathrm{d}x = 0$, since the integrand is identically zero.
- For even $k \geq 2$, $J_{0,k} = \int_0^\pi \sin kx \, \mathrm{d}x = \left[-\frac{1}{k}\cos kx\right]_0^\pi = \frac{1}{k}\left(1-(-1)^k\right) = 0$.

Hence, by induction on $\min(m,n)$, $J_{m,n} = 0$ whenever m, n are both even or both odd.

Or otherwise: substituting $x = \pi - t$ sends $\cos^m x$ to $(-1)^m \cos^m t$ and $\sin nx$ to $-(-1)^n \sin nt$, so $J_{m,n} = (-1)^{m+n+1} J_{m,n}$, which forces $J_{m,n} = 0$ when $m+n$ is even. The integrand is antisymmetric about $x = \frac{\pi}{2}$.

Part (ii)

To bring in cosines, write $\sin^2 x = 1 - \cos^2 x$:

$$\int_0^{\pi/2} \sin^2 x \sin 3x \, \mathrm{d}x = K_{0,3} - K_{2,3}, \qquad K_{m,n} = \int_0^{\pi/2} \cos^m x \sin nx \, \mathrm{d}x.$$

For $m \geq 1$ the boundary term is $-\left[\cos^m x \cos nx\right]_0^{\pi/2} = 0 - (-1) = 1$, because $\cos \frac{\pi}{2} = 0$. So

$$(m+n)K_{m,n} = 1 + mK_{m-1,n-1} \qquad (m, n \geq 1).$$

Starting from $K_{0,1} = \int_0^{\pi/2} \sin x \, \mathrm{d}x = 1$:

$$3K_{1,2} = 1 + K_{0,1} = 2, \qquad K_{1,2} = \tfrac{2}{3},$$
$$5K_{2,3} = 1 + 2K_{1,2} = \tfrac{7}{3}, \qquad K_{2,3} = \tfrac{7}{15}.$$

Also $K_{0,3} = \left[-\frac{1}{3}\cos 3x\right]_0^{\pi/2} = \frac{1}{3}$, since $\cos \frac{3\pi}{2} = 0$. Therefore

$$\int_0^{\pi/2} \sin^2 x \sin 3x \, \mathrm{d}x = \frac{1}{3} - \frac{7}{15} = -\frac{2}{15}.$$

Checks

Part (ii) can be done directly with product-to-sum formulae: $\sin^2 x \sin 3x = \frac{1}{2}\sin 3x - \frac{1}{4}(\sin 5x + \sin x)$, which integrates over $[0, \frac{\pi}{2}]$ to $\frac{1}{6} - \frac{1}{4}\left(\frac{1}{5} + 1\right) = -\frac{2}{15}$. The negative sign is plausible: $\sin 3x < 0$ on $(\frac{\pi}{3}, \frac{\pi}{2})$, exactly where $\sin^2 x$ is largest. Simpson's rule gives -0.133333, and checks the reduction formula and part (i) numerically for many m, n.

Common slips

- Trying to make the substitution $x = \frac{\pi}{2} - t$ in (ii): it turns $\sin^2 x$ into $\cos^2 t$ but $\sin 3x$ into $-\cos 3t$, so the integral is not of the form $I_{m,n}$.
- Forgetting that the reduction formula only applies for $m, n \geq 1$, so the chain in (i) needs its own base cases with an index equal to 0.

Beyond STEP

Reduction formulae like this one are the start of a bigger story: integrals of products of trigonometric functions over a full period are the orthogonality relations behind Fourier series, where symmetry arguments like the one in (i) decide at a glance which coefficients vanish. Boyce and DiPrima, *Elementary Differential Equations*, has a readable introduction to Fourier series and their use in solving differential equations.

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